MID SEMESTER QUESTIONS

student.csv:Age,Gender,StudyHours,Marks,Result,Attendance

1)DATA PREPROCESSING (WITH NULL VALUES IN CSV FILE)

```
import pandas as pd

data = pd.read_csv("student.csv")

print("Dataset:")

print(data)

print("\nDataset Info:")

print(data.info())
```

#Handling Missing Data

```
import pandas as pd

from sklearn.impute import SimpleImputer

data = pd.read_csv("student.csv")

imputer = SimpleImputer(strategy="mean")

data[["StudyHours","Marks"]]=imputer.fit_transform(data[["StudyHours","Marks"]])

print("After handling missing values:")

print(data)
```

2) **LINEAR REGRESSION-HOUSE PRICE PREDICTION (NO NEED TO CREATE CSV FILE)**

```
import numpy as np

from sklearn.linear_model import LinearRegression

sqft = np.array([600, 900, 1200, 1500, 1800]).reshape(-1, 1)

price = np.array([120000, 170000, 220000, 270000, 320000])

model = LinearRegression()

model.fit(sqft, price)

prediction = model.predict([[600]])

print("Slope (m):", model.coef_[0])

print("Intercept (c):", model.intercept_)

print("Predicted Price for 600 Sqft: ₹", int(prediction[0]))
```

3) **K-NEAREST NEIGHBORS CLASSIFICATION (NO NEED TO CREATE CSV FILE)**

```
from sklearn.neighbors import KNeighborsClassifier

X = [[35],[55],[75]]

y = [0,1,1]

model = KNeighborsClassifier(n_neighbors=1)

model.fit(X, y)

print(model.predict([[45]]))
```

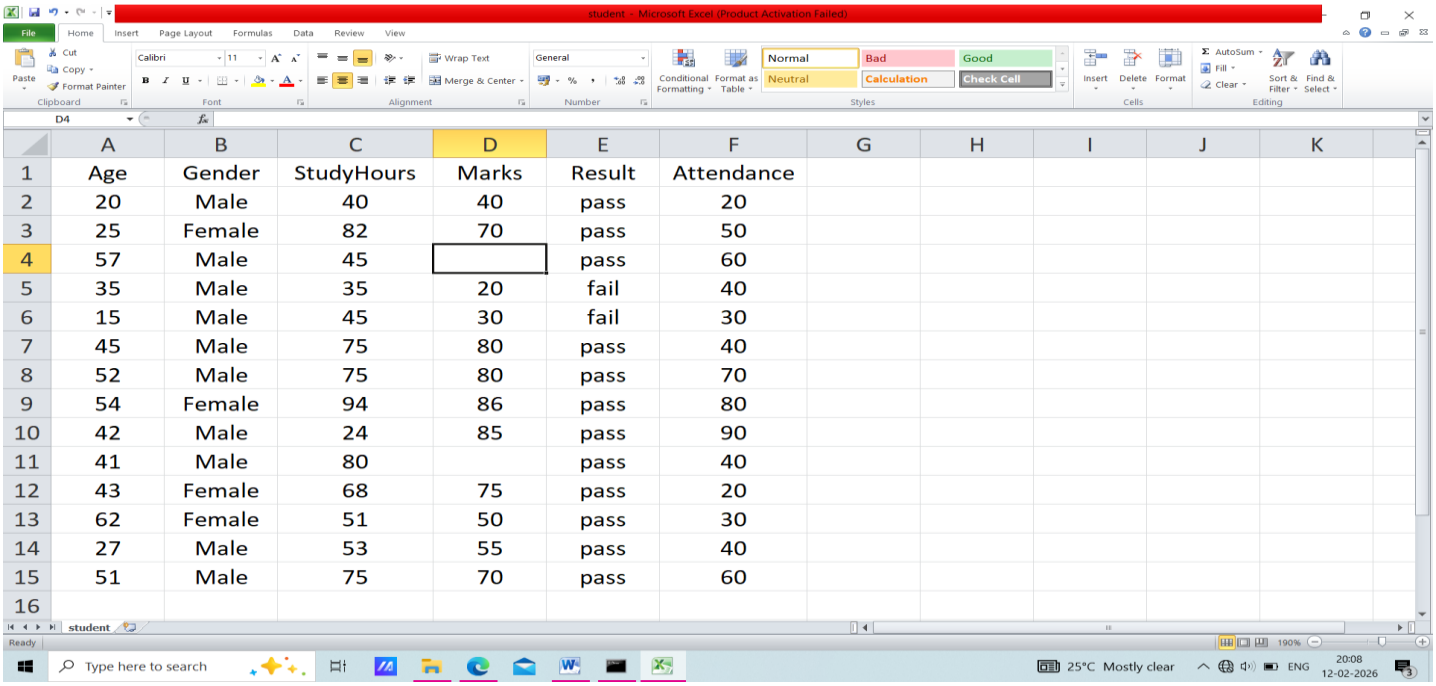
4) IMPLEMENTATION OF SVM USING PYTHON (NO NEED TO CREATE CSV FILE)

```
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score,classification_report,confusion_matrix
iris = datasets.load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
svm_model = SVC(kernel='linear', C=1)
svm_model.fit(X_train, y_train)
y_pred = svm_model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

5)STATISTICAL OPERATIONS USING NUMPY ARRAY (NO NEED TO CREATE CSV FILE)

```
import numpy as np

data = np.array([10, 20, 30, 40, 50])

print("Data:", data)

print("Mean:", np.mean(data))

print("Median:", np.median(data))

print("Minimum value:", np.min(data))

print("Maximum value:", np.max(data))

print("Standard Deviation:", np.std(data))

print("Variance:", np.var(data))
```

NULL VALUES LIKE THIS



	A	B	C	D	E	F
1	Age	Gender	StudyHours	Marks	Result	Attendance
2	20	Male	40	40	pass	20
3	25	Female	82	70	pass	50
4	57	Male	45		pass	60
5	35	Male	35	20	fail	40
6	15	Male	45	30	fail	30
7	45	Male	75	80	pass	40
8	52	Male	75	80	pass	70
9	54	Female	94	86	pass	80
10	42	Male	24	85	pass	90
11	41	Male	80		pass	40
12	43	Female	68	75	pass	20
13	62	Female	51	50	pass	30
14	27	Male	53	55	pass	40
15	51	Male	75	70	pass	60
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